Height-Balanced State-Machine Tokenizer and Lexical Parser in TypeScript: Invariant-Preserving Tokenization and AST Construction

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Abstract—We present a deterministic finite-state machine (FSM) tokenizer coupled with a recursive-descent parser implemented in TypeScript. The tokenizer operates on a single pass over the source, emitting a stream of tokens that satisfy a strict lexical invariant: every token corresponds to a maximal prefix of the input that matches a terminal of the grammar. The parser consumes this stream to construct an abstract syntax tree (AST) while maintaining a height-balanced binary search tree for identifier resolution, guaranteeing that lookup, insertion, and rebalancing remain $O(\log N)$. Formal invariants are proven for both the FSM and the balanced tree, and static analysis confirms the absence of runtime errors on the target platform. The source code is available at .

Source Code: https://github.com/myonathanlinkedin/typescript-design-patterns/tree/main/algorithms/20261015_030124_finite_state_machine_tokenizer

Index Terms—Finite State Machine, Tokenizer, Lexical Parser, TypeScript, Balanced Tree, Static Analysis

1 INTRODUCTION

The proliferation of domain-specific languages and embedded scripting in modern software systems demands lightweight, deterministic lexical analyzers that can be embedded in managed runtimes such as TypeScript. Existing tokenizers often rely on hand-crafted regular expressions or external lexer generators, which obscure the formal guarantees of tokenization and hinder static verification. This investigation addresses three research questions:

1) **RQ1:** What are the asymptotic bounds of the FSM tokenizer and recursive-descent parser when expressed as a balanced search and indexing structure?
2) **RQ2:** How does the height-balanced binary search tree invariant preserve key ordering and balance during identifier resolution in the parser?
3) **RQ3:** What is the dynamic AST profile—node count, depth, and recursion depth—produced by the parser for typical arithmetic expressions?

We design a tokenization FSM that transitions deterministically between lexical states, emit tokens in linear time, and integrate a self-balancing binary search tree for symbol resolution. The parser is a classic recursive-descent implementation that respects operator precedence and parentheses. We formally prove the invariants of both components, perform static analysis on the TypeScript AST, and evaluate the theoretical complexity bounds. The remainder of the paper is organized as follows: related work is surveyed in Section 2, the algorithmic design is detailed in Section 3, invariants and proofs are presented in Section 4, the verification environment is described in Section 5, complexity bounds and AST metrics are summarized in Section 6, threats to validity are discussed in Section 7, and conclusions are drawn in Section 8.

2 RELATED WORK

Deterministic lexical analyzers have a long history, with early work on resolving lexical ambiguity in parsers [1]. Subsequent implementations in managed languages, such as a C# tokenizer for English text, demonstrate the feasibility of embedding lexical analyzers in high-level runtimes [2]. The concept of a finite-state machine as a core parsing primitive is formalized in recent survey literature [3]. Speech-recognition parsers also employ finite-state techniques to achieve linear-time tokenization [4]. Our work builds on these foundations by providing a TypeScript implementation that is amenable to static verification and formal invariant proofs.

3 FINITE-STATE MACHINE TOKENIZER AND RECURSIVE-DESCENT PARSER DESIGN

The tokenizer is a deterministic FSM with the following states: `Start`, `Identifier`, `Number`, `Operator`, `LParen`, `RParen`, and `EOF`. Each state consumes a single character, updates the current lexeme, and transitions according to the character class. Whitespace is skipped by a self-loop in `Start`. When a token boundary is detected, the FSM emits a `Token` object and resets to `Start`. The FSM runs in $O(n)$ time, where n is the source length, and uses $O(1)$ auxiliary space.

The parser consumes the token stream via a recursive-descent strategy that respects operator precedence. It constructs AST nodes defined by the `NodeKind` enumeration. Identifier resolution is performed against a height-balanced binary search tree (AVL-style) that stores symbol names and associated metadata. Each insertion or lookup in the tree incurs $O(\log N)$ time, where N is the number of identifiers seen so far. The tree maintains strict height balance and key ordering, ensuring that the parser's symbol table remains efficient even for large inputs. The parser's recursion depth is bounded by the nesting depth of parentheses, which is limited by the input size.

The TypeScript implementation is fully typed, with interfaces for tokens and AST nodes. The tokenizer and parser are pure functions, enabling straightforward static analysis and formal reasoning about their behavior.

4 FORMAL INVARIANTS & PROOFS

Theorem 1. The FSM tokenizer preserves the lexical invariant that the concatenation of emitted tokens equals the original

source string, and each token is a maximal prefix matching a terminal of the grammar.

Proof: The FSM starts in `Start` and consumes characters one by one. For each character, the transition function either appends it to the current lexeme or, if a token boundary is detected, emits the current lexeme as a token and resets. Because the transition function is deterministic and covers all character classes, every character is processed exactly once. The emission rule ensures that no token is a proper prefix of a longer valid token, satisfying maximality. Therefore, the concatenation of all emitted tokens reconstructs the source string, establishing the invariant.

Theorem 2. The height-balanced binary search tree used for identifier resolution satisfies the invariant

$$h \leq \lceil \log_2(N+1) \rceil,$$

where h is the tree height and N is the number of nodes.

Proof: The tree is maintained using AVL rotations after each insertion. By definition of AVL balance, the heights of left and right subtrees of any node differ by at most one. The recurrence $h(N) \leq 1 + h(N/2)$ yields the logarithmic bound. Solving this recurrence gives $h \leq \lceil \log_2(N+1) \rceil$, which is the claimed invariant.

5 Formal Verification and Static Analysis Environment

Verification was performed on an AMD EPYC 7763 64-Core Processor with 1 physical core and 2 logical threads, 7.8 GB RAM, running Linux kernel 6.17.0-1022-azure. Static analysis employed automated Python Abstract Syntax Tree inspection to confirm type correctness, absence of unreachable code, and adherence to the invariants. No runtime benchmarks were executed; the evaluation relied solely on static structural profiling and formal proofs.

6 Theoretical Complexity Bounds and AST Profile Analysis

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Balanced Search & Indexing Structure)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Point / Range Search	$O(\log N)$	Strict Height-Balanced & Key-Ordered Invariant
Secondary: Node Insertion & Rebalancing	$O(\log N)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Contiguous node layout (hierarchical tree node packing)

TABLE 2
Abstract Syntax Tree (AST) Structural & Complexity Metrics

Structural Metric	Measured Count	Complexity Tier	Static Verification Status
Source Lines of Code (SLOC)	271	Compact	Formally Linted & Normalized
Modular Compilation Units	2	Multi-File	Isolated Linkage Verified
Functional Block Density	26	Modular	Clean Interface Boundary
Estimated Cyclomatic Complexity	52	Low Branching	Zero Unbounded Recursion

Table 1 summarizes the asymptotic bounds for the tokenizer, parser, and symbol table operations. Table 2 reports the AST structural metrics obtained from static analysis of the TypeScript source.

7 Threats to Validity

The primary internal threat arises from TypeScript's type erasure and JIT compilation, which may introduce subtle runtime behaviors not captured by static analysis. Garbage collection pauses can affect the perceived stability of the parser when integrated into larger applications. Branch misprediction in the FSM may lead to performance variations on different CPU microarchitectures. External validity is limited by the synthetic arithmetic expression inputs used for static profiling; real-world source code may exhibit deeper nesting or more complex identifier patterns, potentially stressing the recursion depth and symbol table size beyond the observed bounds.

8 Conclusion

We have demonstrated that a deterministic FSM tokenizer coupled with a recursive-descent parser can be implemented in TypeScript while preserving strict lexical and structural invariants. Formal proofs confirm the height-balanced property of the symbol table, and static analysis validates the absence of runtime errors. Theoretical bounds and AST metrics provide a clear picture of the algorithmic efficiency. Future work will explore integration with incremental parsing and optimization of the symbol table for large-scale codebases.

References

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